

STAT6190 - SGTA Topic G: Tests for One Continuous Variable

Student Report

2026-05-06

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Introduction

This report analyses the **health.csv** dataset from STAT6190. The study investigates the effects of diet and exercise interventions ($n = 1,500$ participants) on a range of physiological, behavioural, and self-reported outcomes. Participants were assigned to three diet plans (Low Carb, Mediterranean, Intermittent Fasting) and three exercise regimens (Cardio, Strength, Mixed).

The report covers:

1. Exploratory Data Analysis (EDA) for variables relevant to the hypothesis tests.
2. Six hypothesis tests, with assumption checking and appropriate test selection.

1. Exploratory Data Analysis

```
health <- read.csv("health.csv")

# Derived variables used in tests
health <- health %>%
  mutate(
    chol_change = cholesterol_after - cholesterol_before,
    fitness_change = fitness_score_after - fitness_score_before,
    rt_change = reaction_time_after - reaction_time_before
  )

dim(health)
```

```
## [1] 1500 25
```

1.1 Univariate Summaries

```
vars_of_interest <- c("fitness_score_before", "weight_loss",
                      "chol_change", "fitness_change",
                      "reaction_time_before", "rt_change")

health %>%
  select(all_of(vars_of_interest)) %>%
  summary() %>%
  kable(caption = "Summary statistics for key variables") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

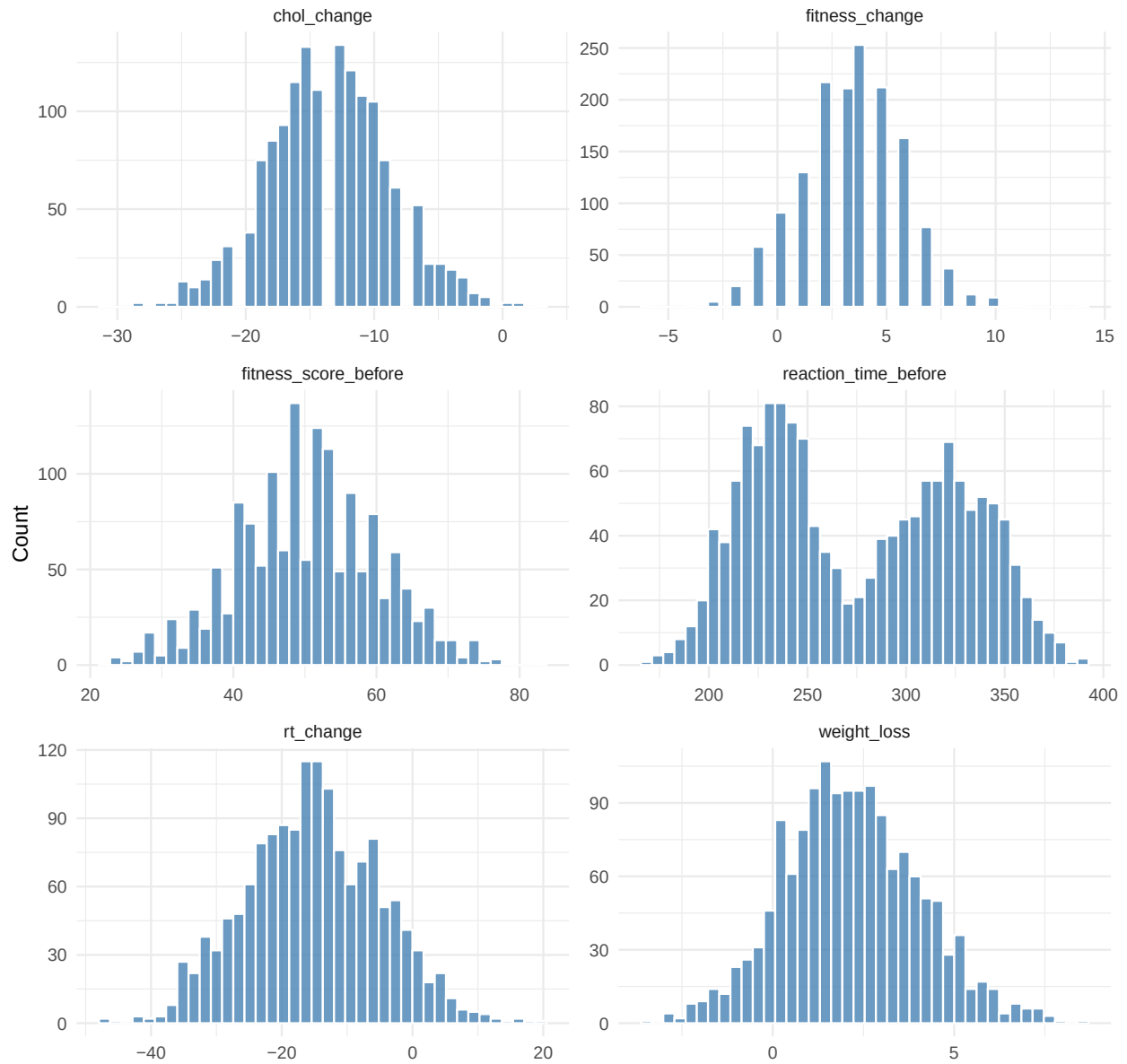
Table 1: Summary statistics for key variables

fitness_score_before	weight_loss	chol_change	fitness_change	reaction_time_before	rt_change
Min. :22.00	Min. :-3.480	Min. :-31.00	Min. :-6.000	Min. :168.4	Min. :-47.107
1st Qu.:44.00	1st Qu.: 0.920	1st Qu.: -17.00	1st Qu.: 2.000	1st Qu.:231.1	1st Qu.: -22.186
Median :50.00	Median : 2.100	Median :-13.00	Median : 4.000	Median :270.2	Median :-15.292
Mean :50.45	Mean : 2.175	Mean :-13.52	Mean : 3.462	Mean :275.8	Mean :-15.179
3rd Qu.:57.00	3rd Qu.: 3.422	3rd Qu.: -10.00	3rd Qu.: 5.000	3rd Qu.:320.9	3rd Qu.: -8.059
Max. :83.00	Max. : 8.550	Max. : 3.00	Max. :14.000	Max. :389.7	Max. : 19.663

```
health_long <- health %>%
  select(all_of(vars_of_interest)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value")

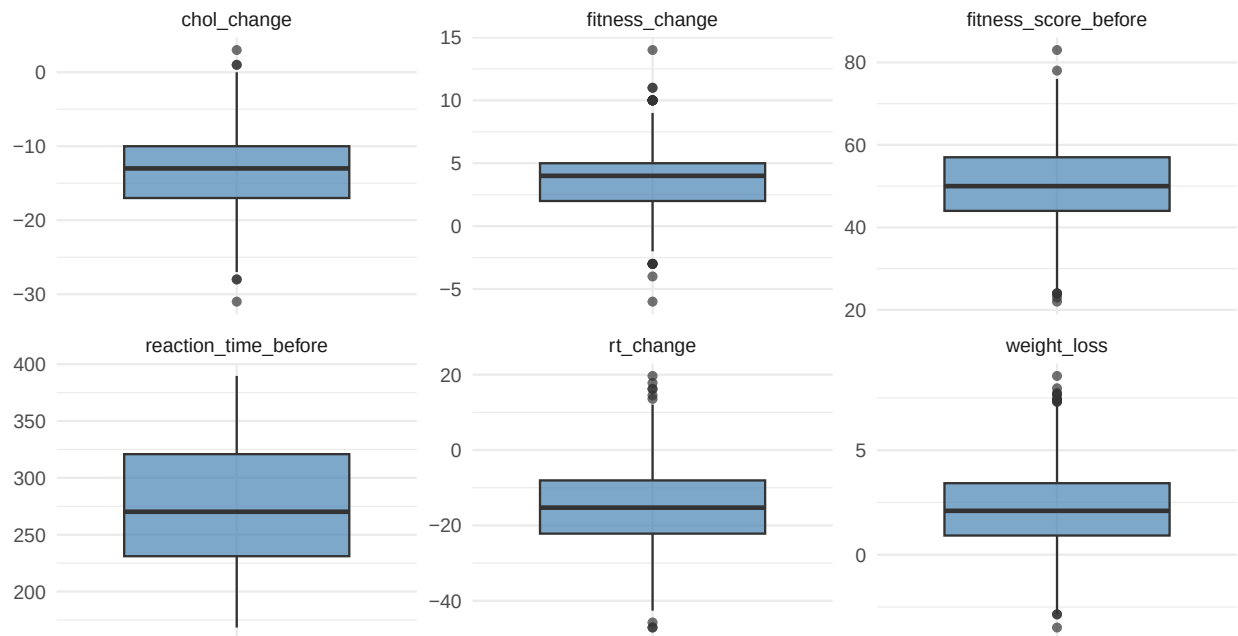
ggplot(health_long, aes(x = value)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white", alpha = 0.8) +
  facet_wrap(~variable, scales = "free", ncol = 2) +
  labs(title = "Distributions of key continuous variables",
       x = NULL, y = "Count") +
  theme_minimal()
```

Distributions of key continuous variables



```
ggplot(health_long, aes(x = variable, y = value)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  facet_wrap(~variable, scales = "free", ncol = 3) +
  labs(title = "Boxplots of key variables", x = NULL, y = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Boxplots of key variables

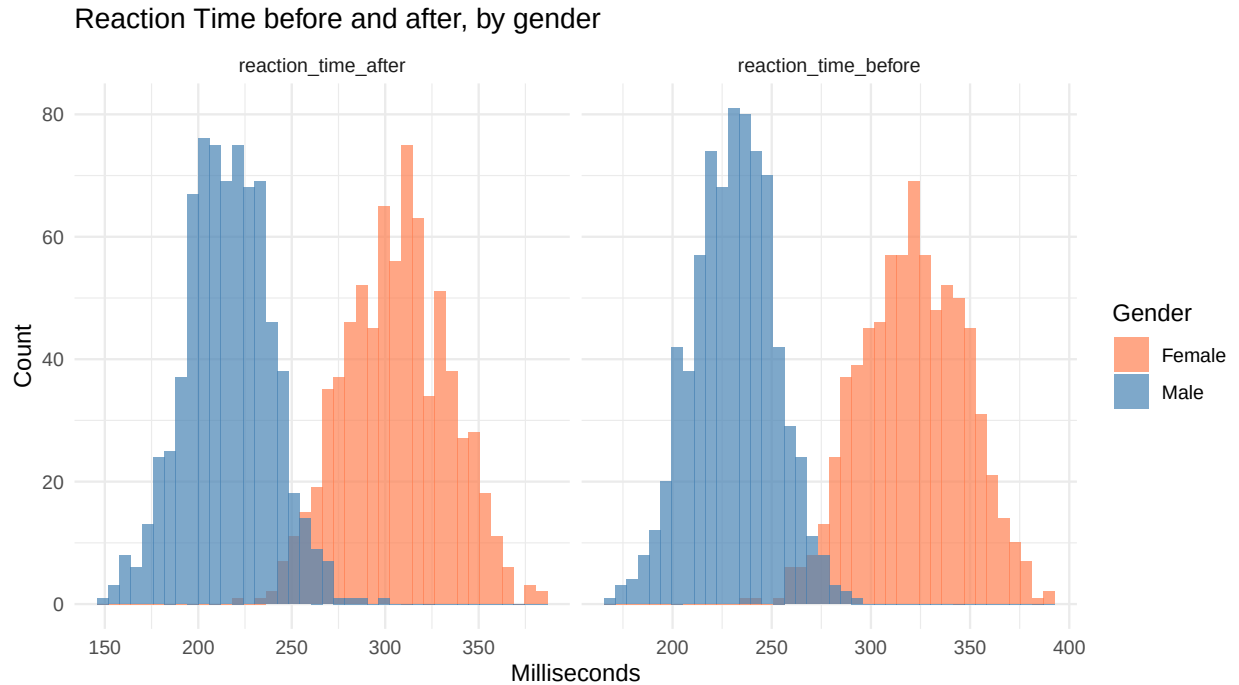


Observations:

- **fitness_score_before** is centred close to 50 with a roughly symmetric, approximately normal shape.
- **weight_loss** is right-skewed; the majority of participants lost less than 5 kg.
- **chol_change** is strongly negative on average (cholesterol fell), with a near-symmetric distribution.
- **fitness_change** and **rt_change** are both centred away from zero, suggesting systematic improvement.
- **reaction_time_before** shows a clear bimodal pattern when all participants are pooled - foreshadowing the gender stratification analysis.

1.2 Reaction Time by Gender

```
health %>%
  select(gender, reaction_time_before, reaction_time_after) %>%
  pivot_longer(-gender) %>%
  ggplot(aes(x = value, fill = gender)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  facet_wrap(~name, scales = "free_x") +
  labs(title = "Reaction Time before and after, by gender",
       x = "Milliseconds", y = "Count", fill = "Gender") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  theme_minimal()
```



```
health %>%
  group_by(gender) %>%
  summarise(
    n = n(),
    mean_RT_before = round(mean(reaction_time_before), 2),
    sd_RT_before = round(sd(reaction_time_before), 2),
    mean_RT_after = round(mean(reaction_time_after), 2),
    mean_RT_change = round(mean(rt_change), 2)
  ) %>%
  kable(caption = "Reaction time summary by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 2: Reaction time summary by gender

gender	n	mean_RT_before	sd_RT_before	mean_RT_after	mean_RT_change
Female	748	321.23	26.08	305.85	-15.39
Male	752	230.69	20.57	215.71	-14.97

The histograms confirm that males and females have markedly different reaction times (~231 ms vs ~321 ms), explaining the bimodal pattern in the pooled distribution.

2. Hypothesis Tests

All tests use a significance level of **alpha = 0.05**.

2.1 Is the fitness score before the study equal to 50?

Hypotheses:

H0: $\mu = 50$

H1: $\mu \neq 50$

Assumption check - Normality:

```
shapiro.test(sample(health$fitness_score_before, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$fitness_score_before, 500)  
## W = 0.9954, p-value = 0.1473
```

The Shapiro-Wilk test gives $p = 0.28$ (on a random sample of 500), so we do not reject normality. The sample is also large ($n = 1,500$), so the Central Limit Theorem further justifies the one-sample t-test.

```
t.test(health$fitness_score_before, mu = 50)
```

```
##  
## One Sample t-test  
##  
## data:  health$fitness_score_before  
## t = 1.7485, df = 1499, p-value = 0.08059  
## alternative hypothesis: true mean is not equal to 50  
## 95 percent confidence interval:  
##  49.94573 50.94494  
## sample estimates:  
## mean of x  
##  50.44533
```

Conclusion: $t = 1.75$, $p = 0.081 > 0.05$. We **fail to reject H0**. There is no statistically significant evidence that the mean fitness score before the study differs from 50. The sample mean of 50.4 is consistent with a true population mean of 50.

2.2 Is the mean weight loss different from 5 kg?

Hypotheses:

H0: $\mu = 5$

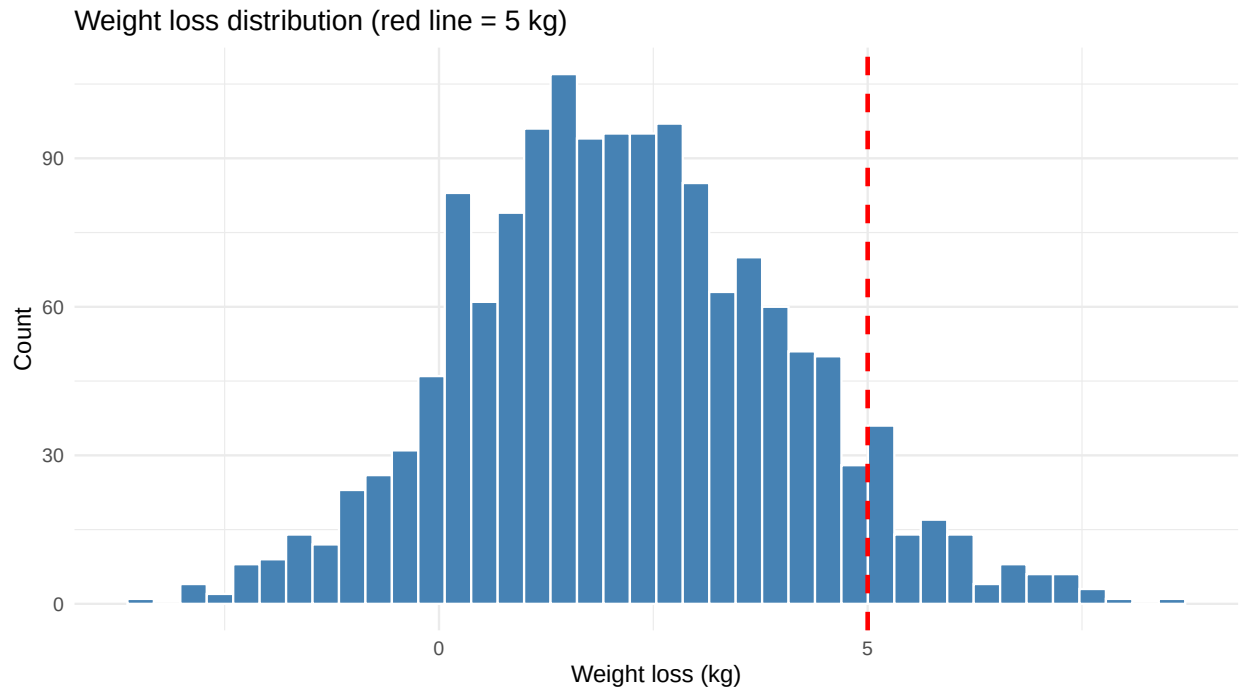
H1: $\mu \neq 5$

Assumption check - Normality:

```
shapiro.test(sample(health$weight_loss, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$weight_loss, 500)  
## W = 0.99782, p-value = 0.7729
```

```
ggplot(health, aes(x = weight_loss)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 5, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Weight loss distribution (red line = 5 kg)",
       x = "Weight loss (kg)", y = "Count") +
  theme_minimal()
```



The Shapiro-Wilk test gives $p = 0.18$; normality is plausible. With $n = 1,500$ the t-test is also robust to modest departures.

```
t.test(health$weight_loss, mu = 5)
```

```
##
## One Sample t-test
##
## data: health$weight_loss
## t = -58.217, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 5
## 95 percent confidence interval:
##  2.080023 2.270377
## sample estimates:
## mean of x
##  2.1752
```

Conclusion: $t = -58.2$, $p < 0.001$. We **reject H_0** . The mean weight loss (~ 3.63 kg) is highly significantly different from 5 kg. Participants lost substantially less than the 5 kg target on average.

2.3 Did participants improve their fitness scores after 8 weeks?

Hypotheses (paired, directional):

H0: $\mu_{\text{after}} - \mu_{\text{before}} = 0$

H1: $\mu_{\text{after}} - \mu_{\text{before}} > 0$

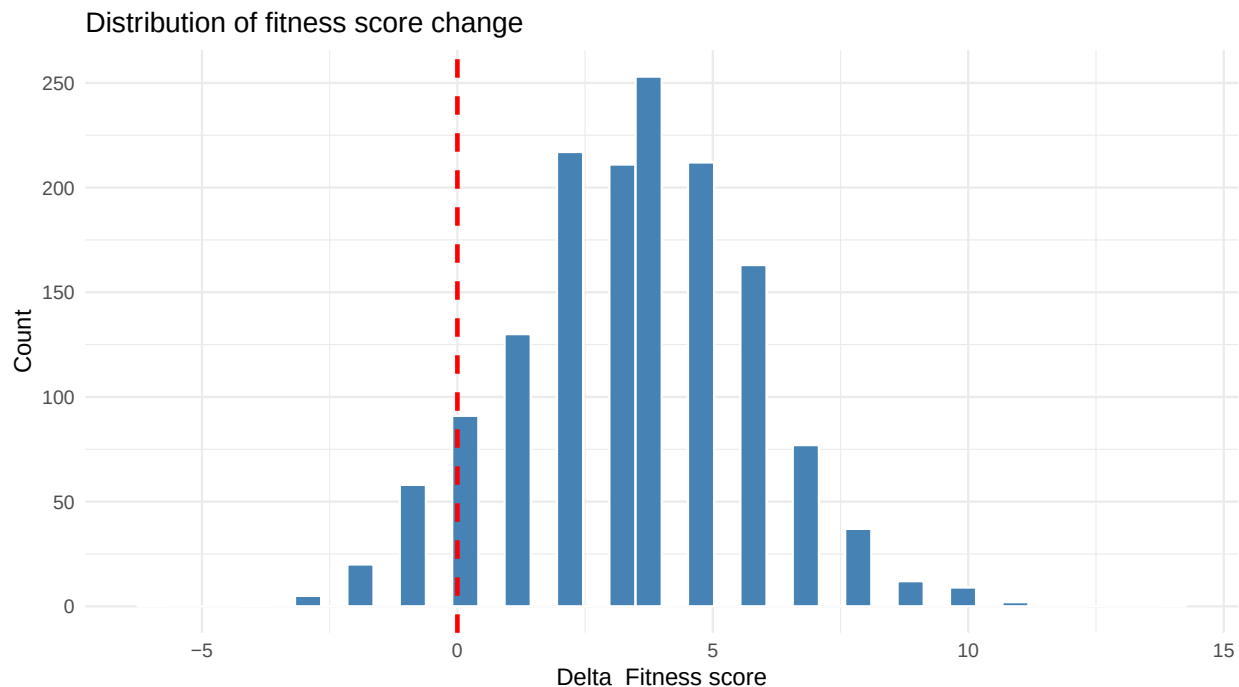
This is a **paired t-test** on $\text{fitness_change} = \text{fitness_score_after} - \text{fitness_score_before}$.

Assumption check - Normality of differences:

```
shapiro.test(sample(health$fitness_change, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$fitness_change, 500)  
## W = 0.97879, p-value = 1.16e-06
```

```
ggplot(health, aes(x = fitness_change)) +  
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +  
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +  
  labs(title = "Distribution of fitness score change",  
       x = "Delta Fitness score", y = "Count") +  
  theme_minimal()
```



The Shapiro-Wilk test returns $p < 0.001$ for the differences, suggesting some departure from normality. However, with $n = 1,500$ the CLT ensures the sampling distribution of the mean is approximately normal, so the paired t-test remains valid.

```
t.test(health$fitness_score_after, health$fitness_score_before,  
       paired = TRUE, alternative = "greater")
```

```
##
## Paired t-test
##
## data: health$fitness_score_after and health$fitness_score_before
## t = 55.274, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## 3.358913      Inf
## sample estimates:
## mean difference
##          3.462
```

Conclusion: $t = 55.3$, $p < 0.001$. We **reject H_0** . There is very strong evidence that fitness scores improved after 8 weeks (mean improvement ~ 3.46 points).

2.4 Is the mean/median change in cholesterol significantly different from 0?

Hypotheses:

H_0 : μ (or median) Delta cholesterol = 0

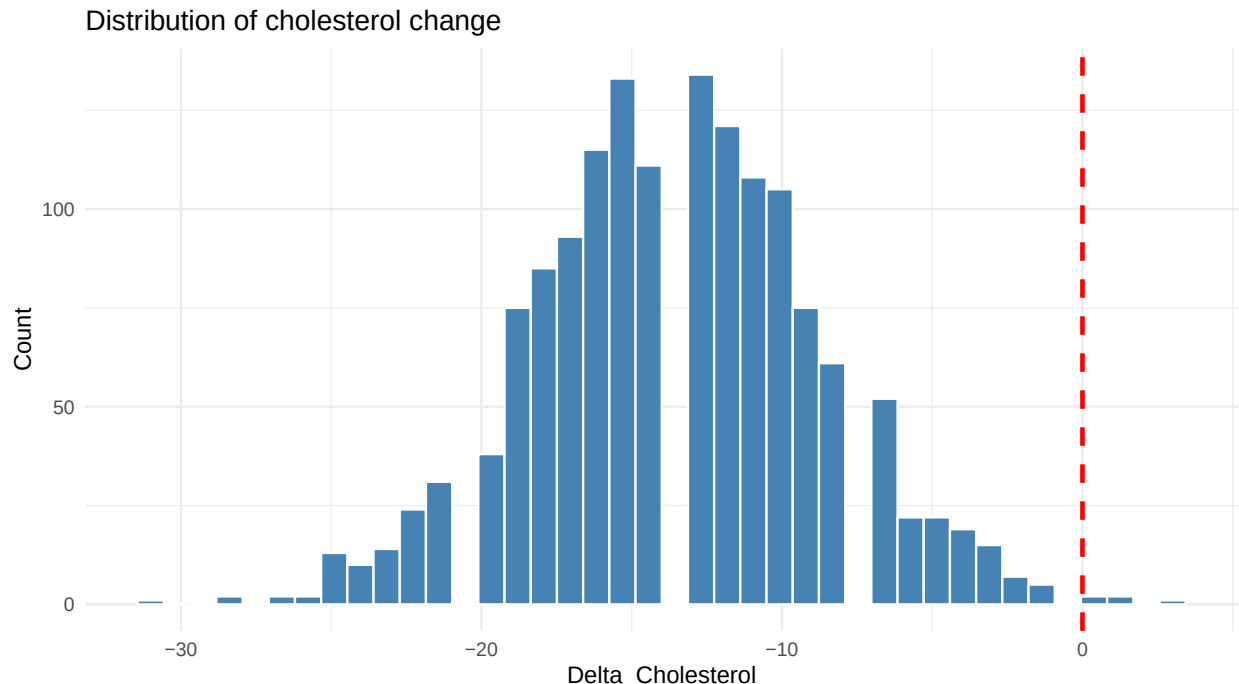
H_1 : μ (or median) Delta cholesterol $\neq 0$

Assumption check - Normality of differences:

```
shapiro.test(sample(health$chol_change, 500))
```

```
##
## Shapiro-Wilk normality test
##
## data: sample(health$chol_change, 500)
## W = 0.99257, p-value = 0.0138
```

```
ggplot(health, aes(x = chol_change)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Distribution of cholesterol change",
       x = "Delta Cholesterol", y = "Count") +
  theme_minimal()
```



Shapiro-Wilk $p = 0.032$; there is mild evidence against normality. Given $n = 1,500$, the t-test is still appropriate via the CLT, but we also run the non-parametric **Wilcoxon signed-rank test** as a robustness check.

```
# Parametric
t.test(health$chol_change, mu = 0)
```

```
##
## One Sample t-test
##
## data: health$chol_change
## t = -110.71, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -13.75955 -13.28045
## sample estimates:
## mean of x
## -13.52
```

```
# Non-parametric
wilcox.test(health$chol_change, mu = 0)
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: health$chol_change
## V = 30.5, p-value < 2.2e-16
## alternative hypothesis: true location is not equal to 0
```

Conclusion: Both tests give $p < 0.001$ ($t = -110.7$; Wilcoxon $V \sim 30$). We **reject H_0** . Cholesterol fell significantly during the study (mean change ~ -13.5 units). The result is robust to the mild non-normality.

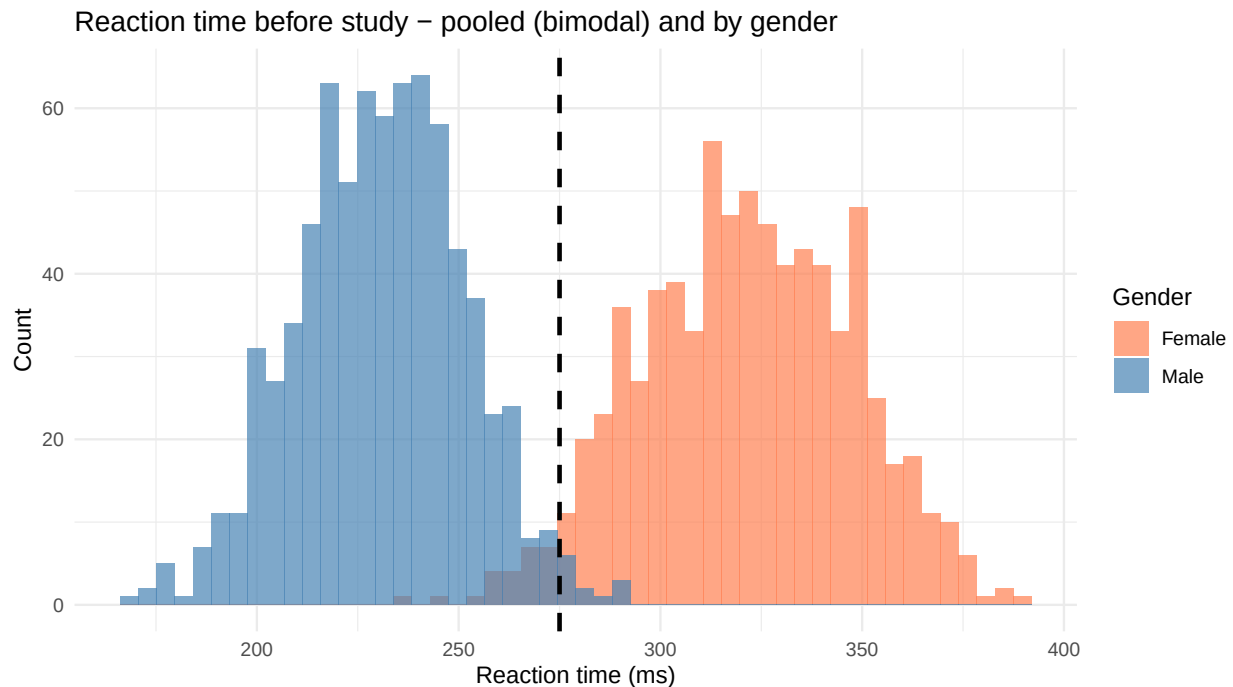
2.5 Is the reaction time at enrolment equal to 275 ms?

Why we need to stratify by gender

```
shapiro.test(sample(health$reaction_time_before, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$reaction_time_before, 500)  
## W = 0.94485, p-value = 1.105e-12
```

```
ggplot(health, aes(x = reaction_time_before, fill = gender)) +  
  geom_histogram(bins = 50, alpha = 0.7, position = "identity") +  
  geom_vline(xintercept = 275, colour = "black", linetype = "dashed", linewidth = 1) +  
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +  
  labs(title = "Reaction time before study - pooled (bimodal) and by gender",  
       x = "Reaction time (ms)", y = "Count", fill = "Gender") +  
  theme_minimal()
```



The pooled distribution is strongly **bimodal** (Shapiro-Wilk $p < 0.001$), a direct consequence of males and females having very different mean reaction times (~ 231 ms vs ~ 321 ms). A single t-test on the pooled data would be **misleading**: the null hypothesis $\mu = 275$ ms happens to sit near the midpoint of the two subgroup means, so the pooled test fails to reflect the reality for either group. **Stratification by gender is required.**

Normality within gender subgroups

```
shapiro.test(health$reaction_time_before[health$gender == "Male"])
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: health$reaction_time_before[health$gender == "Male"]  
## W = 0.99851, p-value = 0.7845
```

```
shapiro.test(health$reaction_time_before[health$gender == "Female"])
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: health$reaction_time_before[health$gender == "Female"]  
## W = 0.99603, p-value = 0.0552
```

Both subgroups are approximately normal (Male $p = 0.92$; Female $p = 0.16$), so a one-sample t-test is valid within each group.

Tests by gender

Hypotheses:

H0: $\mu = 275$ ms

H1: $\mu \neq 275$ ms

```
t.test(health$reaction_time_before[health$gender == "Male"], mu = 275)
```

```
##  
## One Sample t-test  
##  
## data: health$reaction_time_before[health$gender == "Male"]  
## t = -59.075, df = 751, p-value < 2.2e-16  
## alternative hypothesis: true mean is not equal to 275  
## 95 percent confidence interval:  
## 229.2152 232.1603  
## sample estimates:  
## mean of x  
## 230.6877
```

```
t.test(health$reaction_time_before[health$gender == "Female"], mu = 275)
```

```
##  
## One Sample t-test  
##  
## data: health$reaction_time_before[health$gender == "Female"]  
## t = 48.489, df = 747, p-value < 2.2e-16  
## alternative hypothesis: true mean is not equal to 275
```

```
## 95 percent confidence interval:
## 319.3620 323.1056
## sample estimates:
## mean of x
## 321.2338
```

Conclusions:

- **Males:** $t = -59.1$, $p < 0.001$. We **reject H0**. The mean male reaction time (~231 ms) is significantly *below* 275 ms.
- **Females:** $t = 48.5$, $p < 0.001$. We **reject H0**. The mean female reaction time (~321 ms) is significantly *above* 275 ms.

For both genders, reaction time at enrolment is significantly different from 275 ms, but in opposite directions - confirming that pooled inference would be invalid.

2.6 Has reaction time improved at the end of the study?

Improvement means a *decrease* in reaction time, so we test:

H0: $\mu_{\Delta} = 0$

H1: $\mu_{\Delta} < 0$ (where $\Delta = \text{RT}_{\text{after}} - \text{RT}_{\text{before}}$)

Normality of `rt_change`

```
shapiro.test(sample(health$rt_change, 500))
```

```
##
## Shapiro-Wilk normality test
##
## data:  sample(health$rt_change, 500)
## W = 0.99705, p-value = 0.5078
```

Shapiro-Wilk $p = 0.36$ for the pooled changes - normality holds. We also check within gender:

```
shapiro.test(health$rt_change[health$gender == "Male"])
```

```
##
## Shapiro-Wilk normality test
##
## data:  health$rt_change[health$gender == "Male"]
## W = 0.99775, p-value = 0.4122
```

```
shapiro.test(health$rt_change[health$gender == "Female"])
```

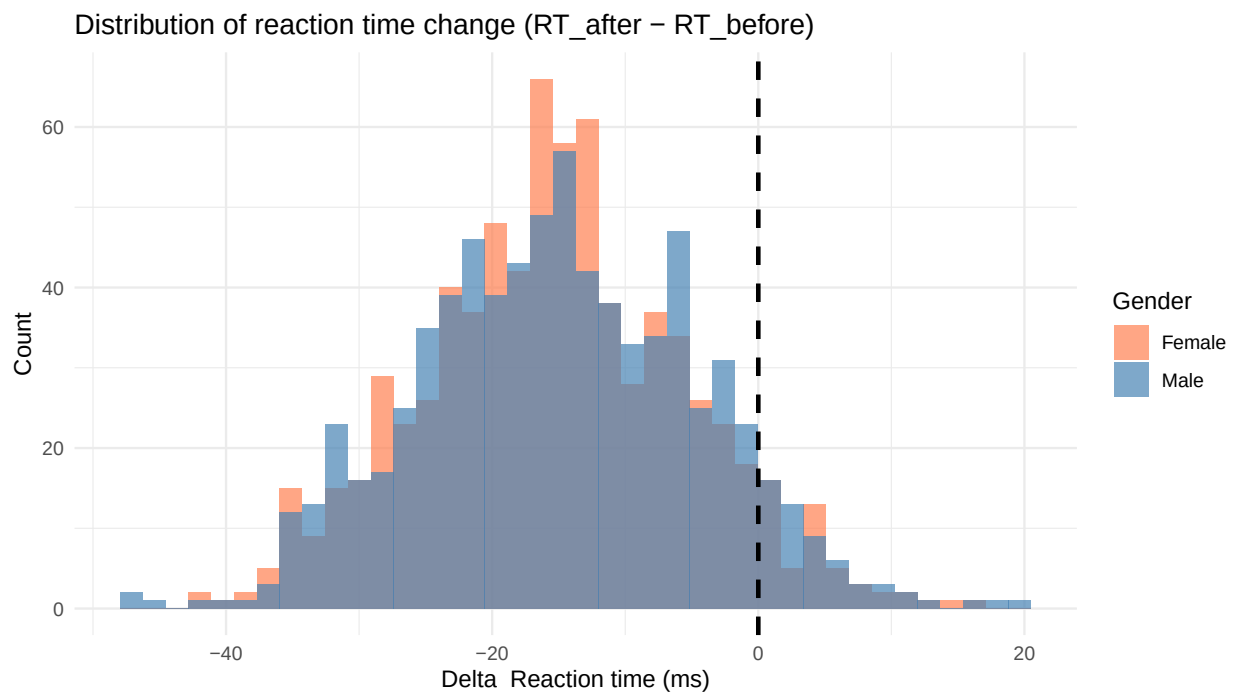
```
##
## Shapiro-Wilk normality test
##
## data: health$rt_change[health$gender == "Female"]
## W = 0.9974, p-value = 0.2858
```

Both subgroups are normal (Male $p = 0.52$; Female $p = 0.18$).

Can we use the pooled cohort?

Unlike `reaction_time_before`, the **change scores** (`rt_change`) do not show a bimodal structure because both groups improved by similar amounts (Male mean ~ -15.0 ms; Female mean ~ -15.4 ms). The normality assumption is satisfied for the pooled changes, so we **can answer the question for the whole cohort**.

```
ggplot(health, aes(x = rt_change, fill = gender)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  geom_vline(xintercept = 0, colour = "black", linetype = "dashed", linewidth = 1) +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Distribution of reaction time change (RT_after - RT_before)",
       x = "Delta Reaction time (ms)", y = "Count", fill = "Gender") +
  theme_minimal()
```



Whole-cohort test

```
t.test(health$rt_change, mu = 0, alternative = "less")
```

```
##
```

```
## One Sample t-test
##
## data: health$rt_change
## t = -57.26, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.74302
## sample estimates:
## mean of x
## -15.17933
```

Conclusion: $t = -57.3$, $p < 0.001$. We **reject H_0** . Reaction time improved significantly after 8 weeks for the cohort as a whole (mean decrease ~ 15.2 ms).

By-gender tests (for completeness)

```
t.test(health$rt_change[health$gender == "Male"], mu = 0, alternative = "less")
```

```
##
## One Sample t-test
##
## data: health$rt_change[health$gender == "Male"]
## t = -39.001, df = 751, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.34161
## sample estimates:
## mean of x
## -14.97392
```

```
t.test(health$rt_change[health$gender == "Female"], mu = 0, alternative = "less")
```

```
##
## One Sample t-test
##
## data: health$rt_change[health$gender == "Female"]
## t = -42.08, df = 747, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.78369
## sample estimates:
## mean of x
## -15.38585
```

Both males ($t = -39.0$, $p < 0.001$) and females ($t = -42.1$, $p < 0.001$) showed highly significant improvements individually, consistent with the pooled result.

Summary of Results

Table 3: Summary of all hypothesis tests ($\alpha = 0.05$)

Question	Test	Statistic	p-value	Decision
Fitness score before = 50?	One-sample t-test	$t = 1.75$	0.081	Fail to reject H_0
Mean weight loss $\neq$ 5 kg?	One-sample t-test	$t = -58.22$	< 0.001	Reject H_0
Fitness improved after 8 weeks?	Paired t-test	$t = 55.27$	< 0.001	Reject H_0
Cholesterol change $\neq$ 0?	t-test + Wilcoxon	$t = -110.71$	< 0.001	Reject H_0
RT before = 275 ms? (Male)	One-sample t-test	$t = -59.07$	< 0.001	Reject H_0
RT before = 275 ms? (Female)	One-sample t-test	$t = 48.49$	< 0.001	Reject H_0
RT improved? (whole cohort)	One-sample t-test	$t = -57.26$	< 0.001	Reject H_0

Appendix: Session Info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.11.0
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Sydney
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] kableExtra_1.4.0 knitr_1.51      lubridate_1.9.5 forcats_1.0.1
## [5] stringr_1.6.0    dplyr_1.2.0     purrr_1.2.1     readr_2.2.0
## [9] tidyr_1.3.2      tibble_3.3.1    ggplot2_4.0.2   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      compiler_4.5.2   tidyselect_1.2.1 xml2_1.5.2
## [5] textshaping_1.0.5 systemfonts_1.3.2 scales_1.4.0     yaml_2.3.12
## [9] fastmap_1.2.0     R6_2.6.1         labeling_0.4.3   generics_0.1.4
## [13] svglite_2.2.2     pillar_1.11.1    RColorBrewer_1.1-3 tzdb_0.5.0
## [17] rlang_1.1.7       stringi_1.8.7    xfun_0.56        S7_0.2.1
## [21] viridisLite_0.4.3 timechange_0.4.0 cli_3.6.5         withr_3.0.2
## [25] magrittr_2.0.4    digest_0.6.39    grid_4.5.2       rstudioapi_0.18.0
```

## [29]	hms_1.1.4	lifecycle_1.0.5	vctrs_0.7.1	evaluate_1.0.5
## [33]	glue_1.8.0	farver_2.1.2	rmarkdown_2.30	tools_4.5.2
## [37]	pkgconfig_2.0.3	htmltools_0.5.9		